Equational reasoning for quantum circuits: from qubits to qudits (and back)

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PhD: equational reasoning for quantum circuits (with a focus on qudits)

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26 March 2026

Roadmap

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1. Quantum circuits in 10 minutes (states, gates, entanglement)
2. Gate sets / fragments: expressiveness + cost model (why we bother)
3. Rewriting & completeness: what an "equational theory" is in practice
4. Qudits: what changes when $d > 2$
5. How my work fits: what I proved, what is ongoing, what is next

Please interrupt me whenever something is unclear.

Quantum computing basics (1/4): states

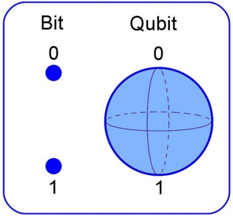
- A **bit** is in $\{0,1\}$. A **qubit** is a *vector* in a 2D complex space.
- Write a (pure) qubit state as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad \text{with} \quad |\alpha|^2 + |\beta|^2 = 1.$$

- Measurement in the computational basis returns:

$$\Pr(0) = |\alpha|^2, \qquad \Pr(1) = |\beta|^2.$$

- New phenomenon vs. classical: **phase** $\Rightarrow$ **interference** (same probabilities, different behaviour in a circuit).



Quantum computing basics (2/4): gates & circuits

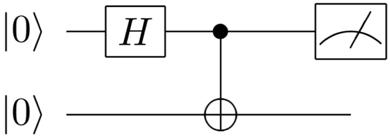
- Gates are **unitary matrices** U (reversible linear maps):

$$U^\dagger U = I, \quad |\psi\rangle \mapsto U|\psi\rangle.$$

- Multi-qubit systems use the tensor product:

$$|\psi\rangle \in (\mathbb{C}^2)^{\otimes n} \;\cong\; \mathbb{C}^{2^n}.$$

- A circuit combines gates by:
  - **Sequential composition** (do one after another)
  - **Parallel composition** (side-by-side)



Quantum computing basics (3/4): phase & interference

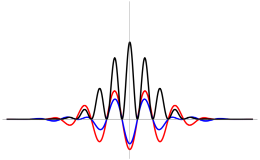
- Two states can have the *same* measurement probabilities but behave differently in a circuit.
- Example (same probabilities, different relative phase):

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

- If you apply a Hadamard H:

$$H|+\rangle = |0\rangle, \qquad H|-\rangle = |1\rangle.$$

- Compiler intuition: **phases control cancellations and reinforcements**.



Quantum computing basics (4/4): entanglement (why 2 qubits $\neq$ 2 bits)

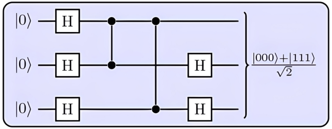
- Two bits have 4 configurations. Two qubits live in a 4D complex space:

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle.$$

- Some states cannot be written as "one qubit $\otimes$ one qubit":

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \quad \text{(entangled)}.$$

- Entanglement is a resource behind quantum advantages
  and it also makes rewriting & verification non-trivial.



Why we rewrite circuits

Three compiler-ish reasons

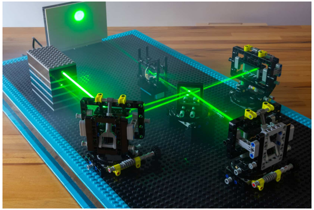
- **Optimisation:** fewer expensive gates, lower depth, fewer ancillas
- **Verification:** prove "my circuit implements the intended unitary"
- **Mapping:** adapt to hardware constraints (connectivity, native gates)

The algebraic viewpoint (what my thesis is about)

- Circuits come with **equations** (rewrite rules) that preserve semantics.
- A good "law book" lets you **transform** circuits *safely* and *automatically*.

Gate sets and fragments: the “LEGO bag” idea

- A **gate set** G = primitives you are allowed to use.
- The **fragment generated by** G = all circuits you can build from G using sequential/parallel composition and wire swaps.
- Example (qubits):
  - Clifford: $\{H, S, \mathrm{CNOT}\}$
  - Clifford+T: Clifford plus one non-Clifford gate T
- Two main questions for a fragment:
  1. **Expressiveness:** what unitaries can you build?
  2. **Reasoning:** do we have a usable complete rewrite system / normal form?

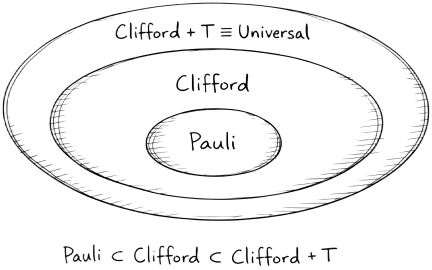


Expressiveness ladder (qubits)

- **Paulis** (X, Z): very limited, but foundational.
- **Clifford** (H, S, CNOT): huge structure, efficiently simulable.
- Add a tiny ingredient:
  - **Clifford+T** becomes (approximately) universal and is a standard FT library.
- A practical slogan:

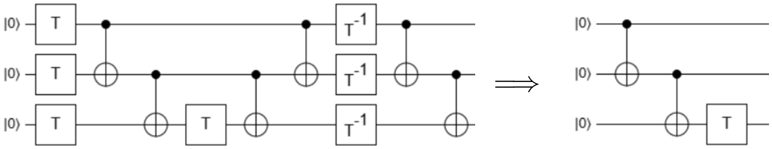
  "Clifford cheap, non-Clifford expensive."

- So we often work in **near-Clifford fragments** to optimise the expensive part.



Cost model intuition: why we obsess over "the hard gates"

- In many fault-tolerant settings, you pay very differently for different gates:
  - Clifford gates: often "easy" / classically trackable
  - non-Clifford gates (like T): often require **magic states** and distillation
- A typical optimiser story:
  1. rewrite to push expensive stuff into a clean shape,
  2. cancel / commute / merge phases,
  3. reduce **T-count**, **T-depth**, or other hardware metrics.
- This is where **complete theories + normal forms** become "compiler technology".



Why fragments matter in practice

Hardware / compilation reality

- Devices expose a **finite menu** of calibrated operations.
- Fault-tolerance makes some gates much more costly (magic states, distillation, etc.).
- Many optimisers use fragment structure: "extract the hard part, normalise it, cancel it".

What we want mathematically

- A **sound** set of local rewrite rules (never changes meaning)
- Ideally **complete**: any true equality is derivable
- Ideally a **normal form**: deterministic canonical circuit shape

State of the art (qubits): a landscape of fragments

Where are we for qubits?

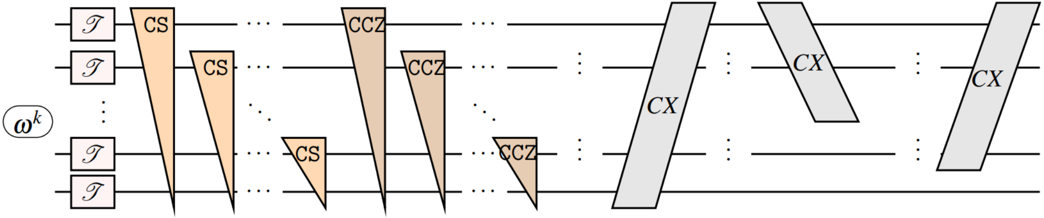
- Many **restricted gate sets** ("fragments") have been studied in detail.
- For a given fragment, people often know:
  - **what it computes** (algebraic / group-theoretic characterisation),
  - whether a **complete equational theory** or **normal form** exists,
  - whether it is **useful for compilers** (good cost model / structure).

Generators	Name	Alternative name	Complete	Minimal	Normal Form
$\mathrm{CNOT} + H + Z$	$\mathrm{Real\ Clifford}_n$	Real stabilizer circuits	☑ [5]	☑ [16]	☑ [5]
$\mathrm{CNOT} + H + Z + \mathrm{CH}$	?	Real stabilizer+CH circuits	☑ [22]	?	?
$\mathrm{CNOT} + H + S$	$\mathcal{C}_n$	Stabilizer / Clifford circuits	☑ [4]	☑ [16]	☑ [4]
$\mathrm{CNOT} + H + T$	$\mathrm{Clifford{+}T}_n$	Clifford+T	~ [6]	~ [16]	~ [6]
$\mathrm{CNOT} + H + P(\theta)$	$\mathrm{U}(2^n)$	Universal	☑ [9]	☑ [9]	?
$\mathrm{Ctrl} + H + s(\theta)$	$\mathrm{U}(2^n)$	Universal with native control	☑ [27]	?	?

A "good case" (qubits): when you avoid Hadamard, structure appears

- Some fragments admit a very **usable algebraic normal form** (great for compilers).
- A common example: **no-Hadamard** fragments built from
  - a **reversible classical layer** (CNOT/X networks on bitstrings),
  - plus a **diagonal phase layer** (phases that depend on parities).
- The diagonal part can be encoded as a **phase polynomial**: rewriting becomes

"move phases around + simplify a polynomial."

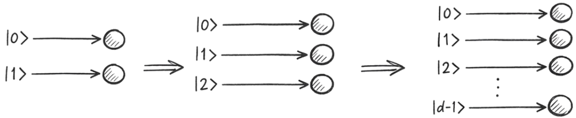


Qudits: what and why

- A **qudit** is a d-level quantum system:

$$|\psi\rangle = \sum_{k=0}^{d-1} \alpha_k |k\rangle, \qquad \sum_k |\alpha_k|^2 = 1.$$

- Why bother?
  - Some platforms are **naturally higher-dimensional** (multiple stable levels).
  - Potential advantages in **compilation** (native gates), **error correction**, or **resource tradeoffs**.
  - Conceptually: many qubit results rely on "dimension 2 coincidences"; we want dimension-robust theory.
- But: moving from $d = 2$ to general d often breaks "nice coincidences".

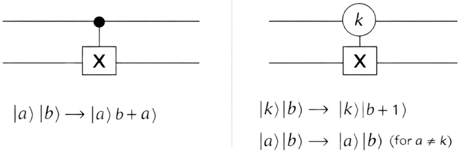


Qudit gates: same spirit, more choices

- Generalised Paulis (one common convention):

$$X_d|k\rangle = |k+1 \bmod d\rangle, \qquad Z_d|k\rangle = \omega^k|k\rangle \;\; (\omega = e^{2\pi i/d}).$$

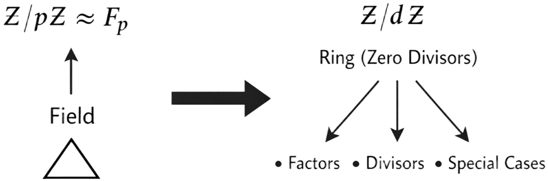
- Entangling gates are where "choices" proliferate:
  - Qubit CNOT is essentially canonical.
  - In higher d, there are multiple reasonable "CNOT-like" gates, and the choice matters for algebra and rewriting.
- Two conventions you may see (schematically):
  - **Controlled-addition (SUM):** $|a, b\rangle \mapsto |a,\, b + a \bmod d\rangle$
  - **Single-value control:** only triggers when the control wire is a chosen basis value (closer in spirit to "binary control")



Prime dimensions: why $d = p$ is an algebra-friendly starting point

- In prime dimension p:
  - $\mathbb{Z}/p\mathbb{Z}$ is a **field** $\mathbb{F}_p$
  - affine maps and polynomials behave cleanly ("degree" works well)
- In composite dimension:
  - $\mathbb{Z}/d\mathbb{Z}$ is a ring with zero-divisors $\Rightarrow$ more case splits
  - many commutation/rewrite arguments become harder to keep local
- So a common strategy is:

start with prime d (clean algebra), then extend.



My thesis question (high-level)

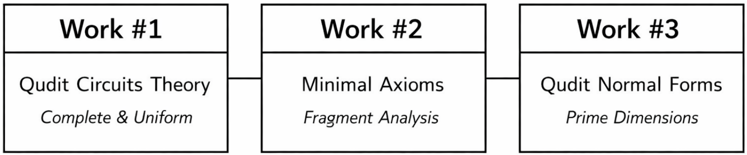
The core question

Can we build small, practical, complete rewrite systems for useful qudit fragments?

- "Useful" = shows up in optimisation / compilation / verification pipelines.
- "Practical" = local rules, ideally bounded arity (e.g. at most 3 wires).

How my work fits (for now)

- **Work #1 (all finite *d*):** a **uniform** complete equational theory for **unitary qudit circuits**, with local (bounded-wire) rules.
- **Work #2 (many fragments): shrink** big axiom sets for near-Clifford fragments and study **minimality / independence**.
- **Work #3 (prime *d*):** generalise the **phase-polynomial / layered normal form & equational theory** story from qubits to prime-dimensional qudits.



How the projects connected (2024 $\to$ 2026)

- **2024 (M2 internship with Simon & Miriam):**
  - Start: "complete theory for qudit circuits with general phases".
  - Main obstacle: too many commutation cases $\Rightarrow$ hard to keep rewriting local and scalable.
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  - $\Rightarrow$ we pivoted to "**remove redundancy / prove minimality**" across fragments $\Rightarrow$ **Work #2**.
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- **2025:** a new abstraction from the Noé & Simon (**Controlled PROPs**) gave the missing piece
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- **2026 (now):** But if we use the other control, some things are smooth, such as phases, which led to **Work #3**.

Work #1 (done): a uniform complete rulebook for qudit circuits

Result (preprint available)

A uniform complete equational theory for unitary qudit circuits (all finite $d \geq 2$)
Available on HAL: `hal-05503323`

- Goal: one **dimension-uniform** theory (same finite schematic rule set for every finite d).
- Key idea (intuition): treat **"control" as a constructor** so that equations stay local.
- Outcome:
  - **Complete** equational theory for unitary semantics,
  - rules are **local** (at most **three wires**),
  - designed to be compatible with **rewrite engines**.
- Take-home message: "I can rewrite qudit circuits like we rewrite qubit circuits, without changing the rulebook every time d changes."

Work #2 (done): smaller presentations for popular fragments

Result (preprint available)

Smaller "law books" for many fragments of quantum circuits
Available on HAL: `hal-05502157`

- Setting: common **near-Clifford** fragments (mostly qubits, plus one qutrit case).
- Problem: completeness is often known, but axiom sets can be **huge** / **redundant**.
- Contribution:
  - Put many fragments in a uniform framework (clean handling of wiring/permutations).
  - Systematically **reduce** presentations and study **minimality / independence**.

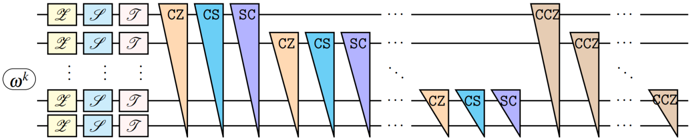
Fragment	Previous Presentation	Our Presentation
Qubit Clifford on n qubits	15 rules [19]	8 rules (minimal for n qubits)
Real Clifford on n qubits	16 rules [17]	10 rules (minimal for n qubits)
Qutrit Clifford on n qutrits	18 rules [15]	10 rules (minimal for 2 qutrits, conjectured for n)
Clifford+T on 2 qubits	18 rules [3]	11 rules (minimal for 1 qubit, simplified for 2)
Clifford+CS on 3 qubits	17 rules [4]	14 rules (minimal for 2 qubits, simplified for 3)
CNOT-dihedral on n qubits	13 rules [1]	11 rules (minimal for n qubits)

Work #3 (ongoing): prime-dimensional phase-affine circuits

In preparation

Completeness for Prime-Dimensional Phase-Affine Circuits

- Goal (prime $d = p$): a qudit version of the **"affine + phase polynomial"** rewriting story.
- Big picture:
  - **Affine skeleton:** reversible affine circuits over $\mathbb{F}_p$ with a **canonical normal form**.
  - **Diagonal phases:** add finite-angle diagonal generators, organised by **polynomial degree**.
  - Derive uniform **transport rules** (move phases through the affine part) and get **unique phase-affine normal forms**.



Next steps (what I want to do next)

- **Polish + extend** the phase-affine work:
  - full version with arbitrary phases. I already have the ruleset and the normal form.
- **Alternative decompositions of controlled gates (prime $d > 3$):**
  - explore decompositions of gates like CS/CZ/CCZ without leaning so heavily on "T-like" resources,
  - e.g. using transpositions X^{ij} (swap $|i\rangle \leftrightarrow |j\rangle$) as a different non-Clifford knob,
  - early evidence: promising for some controlled gates when $d > 3$, but the full story is still open.

Questions & discussion

If you remember only one sentence...

My job is to give compilers and verification tools a small set of rewrite rules that is powerful enough to be ***complete****, mostly beyond qubits, and to explore the world of qudit circuits.*